

Docker Compose (the documented fallback substrate)

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Master technical specification — Volume 6 (Deployment, Tooling & Operations), nano-detail document. **[RES]** research-backed. Scope: the **Docker Compose equivalent** of the HelixVPN gateway stack — the same OCI images as the canonical Podman-rootless substrate, mapped service- by-service to a `compose.yaml`, with all four services (coordinator/helixd, postgres, redis, edge), networks, volumes, secrets, healthchecks, and the capability/security posture. This is the **documented fallback** for environments without Podman — Podman-rootless quadlets remain the §11.4.161 default ([podman-quadlets.md](#)); Compose is the equivalence target for operators on Docker today [05-overview §7.2]. It deepens [05_OV §7.2] with the FACT-grade quadlet→Compose mapping from [research-podman_k8s]. It is a SPEC: the compose file is illustrative of the contract, not the shipping deployment (2–3 refinement passes follow).

Evidence cited inline by id: **[05_OV §N]** = 05-repo-layout-tooling-and-helix-ecosystem.md; **[RPK §N]** = v09-research/research-podman_k8s.md (verified vs latest Docker/Podman docs 2026-06-25 per §11.4.99); **[QUAD §N]** = [podman-quadlets.md](#); **[API §N]** = v03-control-plane/svc-api.md. Unproven facts are marked **UNVERIFIED** per §11.4.6.

0. What this document owns (and does not), and the honest default note

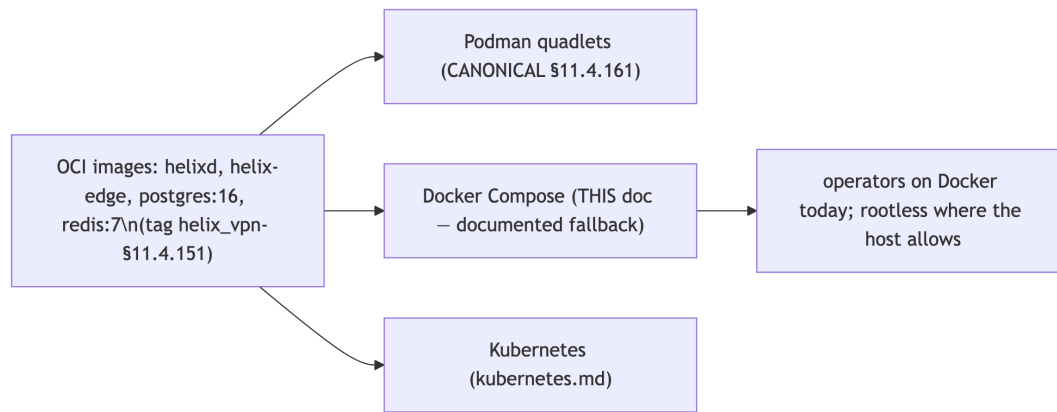
This document owns the **Compose substrate**: the `compose.yaml`, the quadlet→Compose directive mapping, the secrets/volumes/networks/healthchecks, and the rootless-Docker caveats.

It does **not** own: the canonical rootless quadlet substrate ([QUAD], the §11.4.161 default); the K8s manifests ([kubernetes.md](#)); the `helixvpctl deploy compose render` command ([helixvpctl.md](#)); the `env/readyz` contract (canonical in [API §4.7]).

Honest default (§11.4.161/.6). Podman-rootless quadlets are the **canonical, mandated** substrate for HelixVPN (§11.4.161). Compose is the **documented fallback** for operators who already run Docker, NOT a co-equal recommendation. Where the operator's host can run **rootless Docker**, this stack runs rootless; where it can only run **rootful Docker**, that is a host constraint the operator owns — and per §11.4.112 the gap (rootful Docker on this host) is documented honestly, never silently presented as equivalent to the rootless default. A Compose deployment on rootful Docker carries a higher privilege posture than the canonical path, and the operator guide states so plainly.

1. Same images, same env contract — the equivalence target

All three substrates run the **same** OCI images (`helixd`, `helix-edge`, `stock postgres:16` / `redis:7`) with the **same** env contract; Compose loses rootless-by-default but is otherwise a direct mapping of the quadlet pod [05_OV §7, §7.2]. A "single pod" maps to **one Compose project** / **shared network** — Compose has no pod primitive, so inter-service traffic uses the project network (service DNS names like `pg`, `redis`) rather than the pod's `127.0.0.1` [RPK §4].



2. The quadlet → Compose directive mapping (FACT-grade)

The Compose service keys map the quadlet directives one-for-one [RPK §4/§6]:

Concern	Quadlet ([QUAD §3])	Docker Compose	FACT
Add capability	AddCapability=NET_ADMIN NET_RAW	cap_add: [NET_ADMIN, NET_RAW]	[RPK §4/§6]
Drop all caps	DropCapability=ALL	cap_drop: [ALL]	[RPK §4/§6]
Read-only rootfs	ReadOnly=true (+ReadOnlyTmpfs)	read_only: true (+tmpfs: [/run,/tmp])	[RPK §4]
Seccomp	SeccompProfile=<path>	security_opt: ["secomp=<path>"]	[RPK §4]
TUN device	AddDevice=/dev/net/tun:/dev/net/tun	devices: ["/dev/net/tun:/dev/net/tun"]	[RPK §4]
Publish :443/udp	PublishPort=443:443/udp	ports: ["443:443/udp"]	[RPK §4/§6]
Grouping	.pod + Pod=helixvpn.pod	one project / shared network (service DNS)	[RPK §4/§6]
Stateful DB	Volume=helix-pgdata.volume	named volumes:	[RPK §6]
Secrets	podman Secret=	secrets: (file-backed, 0600)	[RPK §4], §11.4.10

Concern	Quadlet ([QUAD §3])	Docker Compose	FACT
Rootless	<code>host ip_unprivileged_port_start=443</code>	same host sysctl (rootless Docker)	[RPK §3/§6]

FACT [RPK §4] — NET_RAW + the host WG module still apply. Exactly as for quadlets ([QUAD §3.2]), the edge needs **NET_RAW in addition to NET_ADMIN**, the **host WireGuard kernel module loaded outside the container**, and `/dev/net/tun` for the userspace path. Docker enables NET_RAW by default for root containers, but the least-privilege baseline here is `cap_drop: [ALL]` then `cap_add: [NET_ADMIN, NET_RAW]` — explicit, not implicit.

Reconciled (§11.4.35, 2026-06-26): the canonical edge capability set is `{NET_ADMIN, NET_RAW}` + `/dev/net/tun` across every Volume-6 substrate (quadlets, this Compose doc, [kubernetes.md](#)) and the [security privesc scan](#). NET_RAW is **CONFIRMED required** by [research-podman_k8s §2] for the **kernel-mode WireGuard fast path** (the edge's primary path); `/dev/net/tun` serves the **userspace boringtun fallback**. The DC4 gate below already asserts exactly this set ("ONLY NET_ADMIN+NET_RAW"); the earlier "ONLY NET_ADMIN" wording in the K8s/overview/security drafts was corrected to match (§11.4.6).

3. The compose.yaml (complete stack)

```
# deploy/compose/compose.yaml (illustrative – the documented
  fallback substrate)
name: helixvpn

services:
  pg:
    image: docker.io/library/postgres:16
    environment:
      POSTGRES_USER: helix_owner          # owner role; helixd
      connects as helix_app (RLS floor) [API §6.5]
      POSTGRES_DB: helix
    secrets: [helix_db_pw]                #
      POSTGRES_PASSWORD_FILE pattern (§3.1)
    cap_drop: [ALL]
    volumes: [helix-pgdata:/var/lib/postgresql/data]
    healthcheck:
      test: ["CMD-SHELL", "pg_isready -U helix_owner -d helix"]
      interval: 5s
      timeout: 3s
      retries: 10

  redis:
```

```

image: docker.io/library/redis:7
cap_drop: [ALL]
read_only: true                                # presence/events are
    Ephemeral (C2 – losing redis loses presence, not identity)
tmpfs: ["/run", "/tmp"]
healthcheck:
  test: ["CMD", "redis-cli", "ping"]
  interval: 5s
  timeout: 3s
  retries: 10

helixd:
image: ghcr.io/helixdevelopment/helixd:helix_vpn-1.0  #
    $11.4.151 prefixed tag
environment:
  DATABASE_URL: postgres://helix_app@pg/helix          # non-
    superuser → RLS enforced [API §6.5]
  REDIS_URL: redis://redis:6379
  PGPASSWORD_FILE: /run/secrets/helix_db_pw            # read
    from the mounted secret (§3.1)
secrets: [helix_db_pw]
read_only: true
tmpfs: ["/run", "/tmp"]
cap_drop: [ALL]                                  #
    control plane needs no extra caps
security_opt: ["no-new-privileges:true"]
healthcheck:
  test: ["CMD", "/helixd", "healthz"]                # →
    GET /readyz semantics [API §4.7]
  interval: 5s
  timeout: 3s
  retries: 10
depends_on:
  pg: { condition: service_healthy }
  redis: { condition: service_healthy }

helix-edge:
image: ghcr.io/helixdevelopment/helix-edge:helix_vpn-1.0
cap_drop: [ALL]
cap_add: [NET_ADMIN, NET_RAW]                    # ONLY
    these two (FACT [RPK §4]); NET_RAW required for WG
devices: ["/dev/net/tun:/dev/net/tun"]            #
    userspace WG (boringtun) fallback path
read_only: true
tmpfs: ["/run", "/tmp"]
security_opt: ["seccomp=../helix-edge-seccomp.json"] #
    tighter than the default profile
ports:
  - "443:443/udp"                                  #
    MASQUE/QUIC ingress (Volume 2)
  - "51820:51820/udp"                              #
    plain-WG ingress
depends_on:
  helixd: { condition: service_healthy }

```

```

secrets:
  helix_db_pw:
    file: ./secrets/db_pw # 0600, gitignored
          (§11.4.10/.30) – NEVER committed

volumes:
  helix-pgdata: {}

```

3.1 Secret delivery (§11.4.10)

Compose secrets are **file-backed**: the raw DB password lives in `./secrets/db_pw` (mode `0600`, parent `0700`, gitignored per §11.4.10/.30), mounted at `/run/secrets/helix_db_pw`. Postgres reads `POSTGRES_PASSWORD_FILE` and helixd reads `PGPASSWORD_FILE` — **the raw value never appears in `compose.yaml`, in `docker inspect`, or in any log** (§11.4.10). `helixvpnctl init --substrate` compose writes `./secrets/db_pw` with the correct perms and a `.gitignore` entry; a pre-store leak audit (§11.4.10.A) runs before any operator-supplied secret is written.

3.2 Healthchecks gate startup ordering

`depends_on: { condition: service_healthy }` makes helixd wait for Postgres + Redis to pass their healthchecks, and helix-edge wait for helixd — the Compose analogue of the quadlet pod's generator-ordered startup ([QUAD §2.2]). The helixd healthcheck maps to the `/readyz` contract (200 only if Postgres `SELECT 1` and Redis `PING` both succeed [API §4.7]), so a half-ready control plane does not accept traffic.

4. Rootless-Docker caveats (FACT-grade)

The same networking caveats that bite rootless Podman ([QUAD §4]) apply to **rootless Docker** [RPK §3/§4]:

Caveat	FACT	HelixVPN handling
:443 rootless bind	Host-level <code>net.ipv4.ip_unprivileged_port_start</code> still governs rootless-Docker privileged binds [RPK §4].	Lower the <code>sysctl</code> (same as quadlets, [QUAD §4]); <code>init</code> prints it as a host prerequisite.
	Compose <code>sysctls</code> : sets per-container kernel params [RPK §4].	Used for edge tunables (e.g.

Caveat	FACT	HelixVPN handling
Per-container sysctls		forwarding) where in-container is correct; host-level binds still need the host sysctl.
Seccomp default	Docker's default profile blocks ~44 syscalls; <code>security_opt: [seccomp=unconfined]</code> disables it [RPK §4].	A custom tighter profile (<code>helix-edge-seccomp.json</code>) is shipped — never unconfined.
No pod 127.0.0.1	Compose has no pod; services reach each other by project-network DNS (pg, redis), not <code>127.0.0.1</code> [RPK §4].	The env contract uses <code>@pg / redis:// redis:6379</code> (service names), unlike the quadlet <code>127.0.0.1</code> .
No auto firewall/NAT	Rootless does not install iptables rules [RPK §3].	Same as quadlets — host-side forwarding or in-container masquerade with <code>NET_ADMIN</code> ; documented prerequisite.

UNVERIFIED [RPK §3]. The pasta 5.8 throughput regression is a Podman-specific note; the rootless-Docker networking stack (slirp4netns / its own) has its own throughput profile — the operator guide says *benchmark the actual host*, never assume parity with the canonical Podman path.

5. The podman kube / Compose bridge (FACT)

Podman can consume a Compose-equivalent K8s YAML directly: `podman kube play` runs a K8s Pod/Deployment YAML, and `podman kube generate` emits one from running containers [RPK §4]. This gives a bridge between the three formats — an operator on Docker Compose can convert to the canonical Podman path, and the K8s manifests ([kubernetes.md](#)) can be replayed under Podman. HelixVPN renders all three from one in-code spec ([QUAD §6]), so the formats are generated-equivalent, not hand-kept-in-sync.

6. helixvpncctl deploy compose render

`helixvpncctl deploy compose --out ./` renders the §3 `compose.yaml` + `secrets/` scaffold from the **same** in-code spec as the quadlets (§11.4.81 cross-platform-parity) [05_OV §6.3, `helixvpncctl` §10]. Parameterized by the `init` inputs (`domain` → edge MASQUE SNI, `image tags` → `helix_vpn-<version>` [§11.4.151], `data-dir` → `secret/volume` paths). A change to the stack shape lands in `quadlet` + `compose` + `kube` together, never in three divergent hand-edited files.

The rendered scaffold:

```
deploy/compose/
├─ compose.yaml                # §3 (the four services)
├─ secrets/
│   └─ db_pw                  # 0600, parent 0700, gitignored
(§11.4.10/.30) – written by init
├─ helix-edge-seccomp.json     # the tighter-than-default seccomp
profile (§4)
└─ .env                       # HELIX_RELEASE_PREFIX + image
tags; gitignored (§11.4.30/.77)
```

The `.env` carries the resolved `helix_vpn-<version>` image tags and the `domain`, so a re-render is idempotent against the same `init` state; `.env.example` (tracked) documents every key per §11.4.77.

7. Operations posture (restart, resource budget, logging, upgrade)

Compose-specific operational rules so the fallback substrate stays within the same host-safety and no-logging guarantees as the canonical path:

7.1 Restart policy

Every service carries `restart: unless-stopped` (the Compose analogue of the quadlet `Restart=always` [QUAD §3]) so a crashed member recovers without operator action, but an operator-initiated `docker compose stop` stays stopped (no fight with intentional maintenance).

```
# applied to every service in §3 (omitted there for brevity):
  restart: unless-stopped
```

7.2 Resource budget (§12.6 — 60% host-memory ceiling)

Per the constitution host-safety ceiling (§12.6 — project procedures **MUST NOT** exceed 60% of total host RAM), each service declares a memory limit so the gateway stack cannot starve the operator's other workloads on a shared host:

```
# per-service (illustrative budget; tuned per host):
  deploy:
    resources:
      limits: { memory: 512M }      # helixd / edge: bounded
      # send queues → bounded RSS [API §9]
      reservations: { memory: 128M }
      # postgres gets the largest share; redis the smallest
      # (presence is tiny)
```

UNVERIFIED: the exact per-service megabyte budgets above are design defaults; they are tuned against the §11.4.169 memory-soak evidence (24 h, RSS slope ≈ 0 [API §13]) at implementation time, not asserted as fixed. The sum **MUST** stay under the §12.6 60% ceiling on the target host.

7.3 Logging (C3 — no traffic, no PII)

Compose logs inherit the control plane's no-logging-by-construction posture [API §4.6, C3]: the access log records route-template + status + latency + tenant + role, never client IP, request body, or auth token. The Compose logging: driver is set to a bounded json-file (size+rotation caps) so logs cannot fill the host disk:

```
logging:
  driver: json-file
  options: { max-size: "10m", max-file: "3" }
```

A `docker compose logs` therefore surfaces operational events, never traffic — the no-logging guarantee is structural (the app emits no traffic log line [API §4.8]), and the bounded driver is the disk-safety belt.

7.4 Upgrade procedure

A version bump is a tag change + recreate, with the §11.4.151 prefixed tags making the release greppable:

```
# 1. bump image tags in .env to the new helix_vpn-<version>
# 2. pull + recreate (Postgres data persists in the named volume):
docker compose pull
docker compose up -d                # recreates changed
                                   services only; pgdata volume retained
# 3. verify:
helixvpncctl status                 # GREEN on the new
                                   images
```

Postgres schema migrations run inside helixd startup (goose, [research-go_cp §6]) against the retained helix-pgdata volume — the upgrade never drops the volume. A rollback re-points .env to the prior helix_vpn-<version> tag and up -d again (the images are immutable, the data persists). Destructive operations on the helix-pgdata volume require the §9.2 hardlinked-backup + operator authorization, never an autonomous down -v.

8. Test points (§11.4.169 comprehensive test-type coverage)

Every PASS cites captured evidence (§11.4.5/.69/.107). Where the test harness boots infra it does so via the containers submodule (§11.4.76); the Compose substrate itself is exercised as a fallback-parity target, never as the primary path.

#	Test type (§11.4.169)	Target	Concrete assertion + evidence
DC1	unit	render determinism	deploy compose twice → byte-identical compose.yaml; image tag carries helix_vpn- prefix
DC2	integration	full-stack compose up	docker compose up → all 4 services healthy (healthchecks pass); helixvpncctl status GREEN
DC3	integration	ordering via healthchecks (§3.2)	helixd starts only after pg+redis healthy; edge only after helixd; captured event order
DC4	security	capability set (§3)	docker inspect shows edge has ONLY NET_ADMIN+NET_RAW;

#	Test type (§11.4.169)	Target	Concrete assertion + evidence
			others none; mutation adding a cap → FAIL
DC5	security	secret not in file/inspect (§3.1)	grep compose.yaml + docker inspect → no raw password; only secret references; perms 0600
DC6	security	seccomp not unconfined (§4)	edge uses the custom profile, never unconfined; mutation to unconfined → FAIL
DC7	e2e	three-substrate parity (§5)	the same image boots under quadlet AND compose; status GREEN on each (recorded §11.4.159)
DC8	chaos	dependency restart	kill pg; assert restart + helixd reconnect; redis loss degrades presence gracefully (C2)
DC9	integration	:443/udp reachable (rootless)	with the host sysctl set, an agent completes a MASQUE handshake to the edge; captured
DC10	meta-test (§1.1)	gates paired	add a cap / inline a secret / set unconfined → each makes its gate FAIL

9. Decision callouts (options + recommendation — §11.4.66/.101)

id	Decision	Options	Recommendation
D-COMPOSE-ROOTLESS	Compose on which Docker mode	(a) rootless Docker; (b) rootful Docker	(a) where the host allows; (b) is an honest §11.4.112-documented higher-privilege gap, not equivalent to the §11.4.161 default
D-COMPOSE-SECRETS	secret delivery	(a) file-backed secrets;; (b) env var	(a) — raw value never in compose.yaml/inspect/log (§11.4.10)
D-COMPOSE-NET	service addressing	(a) project-network DNS; (b)	(a) — Compose has no pod 127.0.0.1; project DNS

id	Decision	Options	Recommendation
		host network	(pg,redis) is the idiomatic + isolated path [RPK §4]

Sources verified

- v09-research/research-podman_k8s.md §4 (quadlet→Compose mapping: cap_drop/cap_add, devices, read_only+tmpfs, security_opt seccomp, ports, single-pod→one-project-network, sysctls, podman kube play/generate bridge), §3 (rootless :443 host sysctl, no auto firewall/NAT), §6 (cross-format mapping table), §2 (NET_RAW + host WG module) — verified vs latest Docker/Podman docs 2026-06-25 per §11.4.99 — [RPK].
- 05-repo-layout-tooling-and-helix-ecosystem.md §7.2 (Docker Compose equivalence target, the docker-compose.yaml sketch, rootless-where-allowed honest gap), §7 (same images/env contract), §6.3 (deploy render via containers submodule) — [05_0V].
- podman-quadlets.md §2/§3/§4 (the canonical substrate this maps from: pod model, edge capability set, rootless caveats) — [QUAD].
- v03-control-plane/svc-api.md §4.7 (/readyz for healthchecks), §6.5 (helix_app non-superuser → RLS), §4.2 (Redis presence ephemeral, C2) — [API].
- Constitution anchors: §11.4.161 (rootless mandate — Compose is the documented fallback, not the default), §11.4.112 (honest documented gap for rootful-Docker hosts), §11.4.10/.30 (file-backed secrets, gitignored, never in compose.yaml/inspect/log), §11.4.76 (containers submodule for test infra), §11.4.81 (one-source three-substrate render), §11.4.6 (no-guessing — UNVERIFIED rootless- Docker throughput mark), §11.4.169 (test-type coverage), §1.1 (paired meta-test mutations).